

Paper outline

Testing LD clusters instead of SNPs — structure, evidence inventory, and gaps

module_sim_LDscnR

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1 The argument in one paragraph

A genome scan tests SNPs; the object of interest is a locus. Benjamini–Hochberg controls the false discovery proportion **among the units tested**, so a single-SNP scan controls it among SNPs while being judged against a locus-level truth. On one simulated panel, 2,467 significant SNPs were **132 distinct LD clusters**, with the rejected set 6.3 times more redundant than the genome as a whole — so “5% FDR over 2,467 discoveries” is a guarantee about a different object from the one being counted. Testing near-independent LD clusters aligns the two. The paper is that argument plus a pipeline that derives its units from the data’s own LD decay rather than from chosen constants.

The central claim requires no simulation. It follows from what BH controls. Everything measured below is supporting evidence for a claim that stands on its own, which is an unusual and exploitable position: the simulations show the size of the effect, not whether it exists.

2 Why it is one paper and not four

Each link is load-bearing rather than merely sequential. Remove any one and the argument fails at a specific point.

Link	What fails without it
LD-decay estimation	The clustering thresholds become chosen constants. The decay fit derives <code>min_r2</code> and the merge distance <i>per chromosome</i> , so a threshold means the same thing on chromosomes with different recombination.
ld_w	No units. ld_w is what makes a cluster a cluster — and, separately, is shown <i>not</i> to work as a filter within them, a negative that earns its place by ruling out the obvious alternative use.
Two-stage reduction	The reported FDR is over SNPs. This is the claim.
Association test	Nothing — and that is the point. EMMAX and LFMM give the same conclusions, which is what shows the result is a property of the units rather than of any test.
Consistency analysis	No guidance on which derived thresholds the answer depends on, so the pipeline is not usable by anyone else.

3 Proposed structure

3.1 1. Introduction

The redundancy problem, stated as an FDR-target mismatch rather than as a multiple-testing complaint. Existing LD-aware scans mostly prune or clump *after* testing; this reverses the order and makes the correction match the unit.

3.2 2. Methods

2.1 Decay estimation and the derived thresholds. 2.2 ld_w . 2.3 Two-stage complexity reduction. 2.4 Cluster-level testing — **Simes over members**, which beat every single-marker summary tested. 2.5 The consistency analysis.

3.3 3. Results

Section	Headline	Status
3.1 The redundancy problem	2,467 significant SNPs are 132 clusters; small- p tail $6.3\times$ more redundant than the genome	done
3.2 What clustering buys	Precision $2\times$ higher (62 of 71 panels). A median 51 of 67 single-SNP flags tag no QTN, against 7 of 10	done
3.3 How to summarise a cluster	Simes $>$ eMLG $>$ representative $>$ min- p $>$ highest- ld_w	done
3.4 The metric is a claim	F_β crossover at $\beta \approx 2$; below it clustering wins, above it does not	done
3.5 Engine replication	EMMAX and LFMM: same composition result, same crossover	done
3.6 Sensitivity	Two axes — selection (27-cell grid) and test (engine) — reported separately	not run
3.7 Empirical demonstration	—	missing

3.4 4. What does not work

Short, and it earns its place by ruling out the obvious things a reader will propose. The C-score as a ranking statistic; four filters (ld_w top- k , MAF, cluster size, geometric exclusion) all strongly enriched for QTN-bearing units and none improving precision-recall. **Enrichment predicts precision within a restriction, but no restriction beats using every cluster** — the recall lost always exceeds the precision gained.

3.5 5. Limitations

4 Evidence inventory: what would a referee ask for?

Gap 1 — no empirical demonstration. Everything is simulated. The natural companion is the 3sp panel, but its support statistic currently measures the *filter* rather than the test (units the filter selected but the test did not reject overlap peaks at 69.2% against the discoveries' 78.2%, $p = 0.18$), so it cannot carry a validation claim as it stands. **This is the largest gap.**

Gap 2 — the advantage is concentrated in one simulation cell. Simes beats the single-SNP scan in 19 of 19 panels in V0.5_c1 and the paired tests in the other three cells are **null, not adverse** (26 W / 22 L, all $p > 0.4$) — several panels are ties in which neither analysis recovers anything. Must be stated in the Results, not a supplement: *decisive where there is signal, indistinguishable where there is not.*

Gap 3 — the sensitivity analysis is planned, not run. It is also the link that makes the pipeline usable by someone else. Design is in TODO_sensitivity.md: $3 \times 3 \times 3$ over `ld_w_threshold`, `min_r2_rho` and minimum cluster size (9 re-clusterings, not 27, since the size floor is a post-filter), reported on two axes with no threshold recommended.

Gap 4 — one simulation generation. Re-clustering at the newer decay settings changes nothing ($p = 0.25$ and 0.85), so this is closed for the decay parameter — but all results come from one simulator under one demographic model.

5 Claims the paper should not make

- That clustering finds more loci. It finds **fewer** and is right more often.
- That ld_w filtering helps. Four variants, none did.
- That any performance number transfers. The **ratios** are the transferable part; absolute precision is a property of this signal-to-noise regime.
- That a threshold is recommended for the consistency score. Every threshold tried on a C-like quantity failed, including the one that looked principled.
- That enrichment for containing a QTN is irrelevant to detection. It predicts precision; it just cannot be exploited by restriction.

6 Two methodological findings that may be worth their own space

Name the axis a stability measure is on. A consistency score computed over parameters that move only the selection measures how stable the *selection* is. "Called under every setting" reads as a claim about the finding. The second association engine is the missing test-side axis, which makes the engine sweep and the sensitivity analysis one design rather than two.

Replicate specifically where nobody doubts the result. The LFMM replication on the panel side surfaced two sampler defects that had been sitting in the original arm for a day with a manuscript being written around them. A replication that *agrees* leaves such defects in place; the disagreement is the diagnostic.

7 Provenance

Result document: `ld_clustering_result.pdf`. Full record including everything withdrawn: `development_history.pdf`. Both build from committed CSVs. Branch `null-rework-two-basis`.